

B. Array Walk

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You are given an array $a_1, a_2, \dots, a_n$, consisting of n **positive** integers.

Initially you are standing at index 1 and have a score equal to a_1 . You can perform two kinds of moves:

1. move right — go from your current index x to $x + 1$ and add a_{x+1} to your score. This move can only be performed if $x < n$.
2. move left — go from your current index x to $x - 1$ and add a_{x-1} to your score. This move can only be performed if $x > 1$. **Also, you can't perform two or more moves to the left in a row.**

You want to perform **exactly** k moves. Also, there should be no more than z moves to the left among them.

What is the maximum score you can achieve?

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of testcases.

The first line of each testcase contains three integers n, k and z ($2 \leq n \leq 10^5$, $1 \leq k \leq n - 1$, $0 \leq z \leq \min(5, k)$) — the number of elements in the array, the total number of moves you should perform and the maximum number of moves to the left you can perform.

The second line of each testcase contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^4$) — the given array.

The sum of n over all testcases does not exceed $3 \cdot 10^5$.

Output

Print t integers — for each testcase output the maximum score you can achieve if you make exactly k moves in total, no more than z of them are to the left and there are no two or more moves to the left in a row.

Example

input	Copy
4 5 4 0 1 5 4 3 2 5 4 1 1 5 4 3 2 5 4 4 10 20 30 40 50 10 7 3 4 6 8 2 9 9 7 4 10 9	
output	Copy
15 19 150 56	

Note

Educational Codeforces Round 92 (Rated for Div. 2)

Finished

Practice



→ Virtual participation

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Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: PyPy 3.10 (7.3.15, 64bit)

Choose file: Browse... No file selected.

Submit

→ Problem tags

brute force dp greedy *1600

No tag edit access

→ Contest materials

- Announcement
- Tutorial (en)

In the first testcase you are not allowed to move left at all. So you make four moves to the right and obtain the score $a_1 + a_2 + a_3 + a_4 + a_5$.

In the second example you can move one time to the left. So we can follow these moves: right, right, left, right. The score will be $a_1 + a_2 + a_3 + a_2 + a_3$.

In the third example you can move four times to the left but it's not optimal anyway, you can just move four times to the right and obtain the score $a_1 + a_2 + a_3 + a_4 + a_5$.

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